

Online Fair Allocation of Indivisible Goods: A No-Regret Learning Approach to Balancing Fairness and Efficiency

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1 Background and Motivation of This Project

Motivation

GAIW (Games, Agents and Incentives Workshop) is an important workshop under AAMAS focusing on game theory, mechanism design, and computational social choice. The 2024 edition was its sixth installment, and we were delighted that our paper¹ was selected as one of the 26 accepted papers. Although GAIW has no proceedings and is non-archival, selection is nevertheless highly competitive, as evidenced by the program committee and the authors of accepted papers. For example, Paul W. Goldberg, a leading scholar in game theory and algorithms, was involved, and well-known researchers such as Inbal Talgam-Cohen and David Parkes were also closely connected with the workshop.

Participating in this event greatly broadened my research perspective and strengthened my motivation. My strongest impression was that many traditional theoretical computer science (TCS) researchers have entered game-theoretic research, and both algorithmic and mechanism-design work in this area are flourishing. In Taiwan, these topics may receive comparatively less attention. Regarding online algorithms and envy-free allocation, I had previously thought of them as classical problems (for example, cake cutting), but I was surprised to see how many variants and algorithmic directions remain active and continue to advance. Moreover, in addition to TCS conferences such as ICALP, SODA, STOC, and FOCS, game theory and computational social choice papers are now published extensively in major AI venues such as IJCAI, NeurIPS, AAI, and AAMAS. Through this project, I hope to build on the research momentum accumulated so far, study important algorithm-design questions in depth, and ultimately publish in these important TCS/AGT/ML/COMSOC conferences², while also creating more opportunities for international collaboration and exchange.

¹An extension of the outcome of an earlier project (MOST 110-2222-E-032-002-MY2), which studied the existence of pure Nash equilibria and upper/lower bounds on the price of anarchy for three or more political parties.

²TCS: theoretical computer science; AGT: algorithmic game theory; ML: machine learning; COMSOC: computational social choice.

Over the past two years, we have carried out a Taiwan–France Orchid Project (NSTC 113(114)-2927-I-032-501) in close collaboration with Prof. Guillaume Fertin at Nantes University. We have already obtained preliminary results on envy elimination for indivisible goods. Building on the problem considered by Bentert et al. [10], where agents already hold partially allocated goods and one tries to eliminate envy by adding goods of nonnegative integer value and limited supply³, we proved that the problem remains NP-complete even for exactly two agents. When there is only one type of additional good available (though agents may value it differently), there exists a polynomial-time algorithm for finding a valid allocation. We plan to extend this result and publish it in the future.

1.1 Problem Background

Inspired by the Taiwan–France Orchid Project, we would like to investigate *online* fair allocation of indivisible goods in greater depth. The core problem of this proposal is the following:

Given n agents and m types of indivisible goods, suppose that each agent’s value for a bundle is an additive valuation or, more generally, a monotone valuation. In each round, only one item is revealed, and it must be allocated to an agent immediately. Under a fairness criterion (such as envy-freeness), we seek, after T rounds, an allocation that maximizes utilitarian social welfare or Nash social welfare while satisfying approximate fairness or vanishing unfairness.

We further ask:

- **When future information about items is predictable, how does that affect the optimal solution to the above problem? How does the achieved social welfare depend on prediction accuracy?**

Finally, we will challenge the following harder core questions:

- **If in each round we may decide whether the current item should be donated to charity, and after the final T rounds the value of the charity bundle to each agent does not exceed the value of that agent’s own allocation, does there exist an online algorithm that computes an approximate allocation satisfying Envy-Freeness up to Any Good (EFX)?**
- **Can the degree of EFX violation be expressed by a suitable potential function, so that the existence of an EFX allocation can be proved via a dynamic algorithm that decreases that potential?**

Below we briefly review the background on the possibility of fair allocation and the design of allocation algorithms.

³If the supply of additional goods is unlimited, then one can decide in polynomial time whether there exists an additional allocation that eliminates envy.

Divisible goods. Given a set M of m goods, distributing them “fairly” among a set N of n agents has long been a classical problem in mathematics, computer science, and economics. Divisible goods, such as in cake cutting, have been extensively studied. Both maximum Nash social welfare (see Definition 5) and envy-free allocations (see Definition 1 with $\alpha = 1$) are known to admit polynomial-time algorithms in the divisible setting [31]. The classical “I cut, you choose” mechanism is a familiar example for two agents dividing a single good. The idea of envy can be traced back at least to Foley’s 1966 dissertation [22]. Informally, an allocation is envy-free if, after the goods are allocated, every agent evaluates every other agent’s bundle as no better than her own. Most discussions of fair allocation focus on envy-freeness and proportionality; the latter is often related to the maximin share guarantee. One may also seek to maximize Nash social welfare (NSW), whose objective is the geometric mean of agents’ utilities. Compared with utilitarian social welfare (USW), which maximizes the sum of utilities, NSW better captures fairness. For example, if one agent’s utility is extremely small (say close to 0), then the NSW value is also very small, whereas USW may remain large. Moreover, maximizing NSW comes with multiple fairness guarantees [16]. In the divisible setting, maximum NSW can be computed via Eisenberg–Gale convex programming [19].

Indivisible goods. When goods are indivisible, a “fair” allocation may fail to exist, and even when one exists, computing it is not necessarily polynomial-time tractable. Regarding existence, it is easy to see that envy-free allocations for indivisible goods need not exist. For example, if there is only one item and at least two agents, then as long as every agent values the item positively, any agent who does not receive it will envy the winner. Regarding computational complexity, given $m \geq 1$ indivisible goods, deciding whether there exists an envy-free (EF) allocation for $n \geq 2$ agents is **NP**-complete [12]. A simple special case already illustrates this fact: when $n = 2$ and both agents have identical valuations, the problem reduces to the classic Two-Partition problem. Likewise, when $n = 2$, an envy-free allocation corresponds to a maximin-share fair allocation, so deciding whether an MMS allocation exists is also **NP**-complete. More generally, even computing an agent’s MMS value is **NP**-hard [33]. In addition, maximizing Nash social welfare for indivisible goods is **APX**-hard [25], and the best currently known polynomial-time approximation ratio is $e^{1/e} \approx 1.445$ [18].

Relaxations of envy-freeness. Envy-Freeness up to One Good (EF1) is a relaxation of EF (see Definition 2) introduced by Lipton et al. in 2004 [26]. Informally, an allocation satisfies EF1 if every instance of envy can be eliminated by removing some single item from the envied agent’s bundle. EF1 allocations always exist and can be found in polynomial time. A well-known fact is that if one can compute an allocation maximizing Nash social welfare, then that allocation is both EF1 and Pareto optimal (PO). Envy-Freeness up to Any Good (EFX) [16] is another relaxation of EF, but it is stricter than EF1: after removing **any** one item from the envied agent’s bundle, no agent should still envy that bundle. If we denote by Π_{EF} , Π_{EF1} , and Π_{EFX} the sets of allocations satisfying EF, EF1, and EFX, respectively, then clearly $\Pi_{\text{EF}} \subseteq$

$\Pi_{\text{EFX}} \subseteq \Pi_{\text{EF1}}$. Whether EFX allocations always exist for general m, n remains a major open problem in economics [16]. Nevertheless, approximate EFX results are known: after removing any item from agent i 's bundle, every other agent j values her own bundle at least an α fraction of the remaining value of i 's bundle, for some $\alpha \in (0, 1]$. Better approximate EFX guarantees are also known when one is allowed to donate a sublinear number of goods to charity, under the condition that no agent envies the charity bundle [1, 17]. Interestingly, the upper bounds on the number of donated goods are derived using tools from extremal graph theory.

Online fair allocation of indivisible goods. We have seen that even in the offline setting, where all indivisible goods are present and all valuations are known, allocations satisfying a target fairness property may fail to exist, and the associated computational problems are often difficult; the existence of EFX itself remains open. It is therefore natural to expect that fair allocation becomes even more challenging in the online setting, where information is incomplete. Online fair allocation arises from applications such as food-bank distribution, allocation of donated medical supplies, and organ matching, where resources are time-sensitive and must be allocated immediately. In recent years, online fair allocation has attracted growing attention at major AI and TCS conferences [2, 8, 23, 29, 32, 35]. The online setting differs substantially from the offline one: items arrive one by one (or in small batches), and each revealed item must be assigned irrevocably to an agent at once. Thus, traditional offline procedures such as round robin are no longer directly applicable, because when it is an agent's turn, she cannot choose her favorite item from the remaining pool—only the currently revealed item can be accepted or rejected.

Observation 1. *For online allocation of indivisible goods, no online algorithm can guarantee EF1 on all instances.*

Proof. Fix any online algorithm **ALG**. We consider two agents with nonnegative additive valuations. Suppose there are two agents A, B and at most three items g_1, g_2, g_3 arriving sequentially. Recall that EF1 requires that for every $i, j \in \{A, B\}$ there exists some good $g \in \mathcal{A}_j$ such that $v_i(\mathcal{A}_i) \geq v_i(\mathcal{A}_j \setminus \{g\})$, where $\mathcal{A}_i$ is the bundle allocated to agent i . Since we are in the online setting, we assume that **ALG** does not know the total number of rounds in advance.

Case 1: suppose the first two items satisfy

$$v_A(\{g_1\}) = v_B(\{g_1\}) = 1, \quad v_A(\{g_2\}) = v_B(\{g_2\}) = 1.$$

Then **ALG** must make one of the following two decisions on g_1, g_2 : (i) allocate both items to the same agent (call this *stacking*), or (ii) give one item to each agent (call this *splitting*). Since we may act as an adversary who knows how the algorithm behaves, if **ALG** stacks the first two items, then the adversary simply terminates the instance after two rounds. In that case, $v_A(\mathcal{A}_A) = 2$ and $v_B(\mathcal{A}_B) = 0$, and even after removing either item from $\mathcal{A}_A$, agent B still

envies agent A , so EF1 fails. In fact, even if a third item arrives and we set $v_A(\{g_3\}) = 2$ and $v_B(\{g_3\}) = 0$, EF1 still cannot be satisfied regardless of how g_3 is allocated.

From the above observation, when the total number of items is unknown, any algorithm that aims to maintain EF1 must choose *splitting* when the first two items arrive. We therefore construct the following instance.

Case 2: consider the following valuations, where $\varepsilon \in (0, 1)$ is a fixed small constant and g_1, g_2, g_3 arrive in order:

	g_1	g_2	g_3
A	ε	2ε	100
B	2	1	100

Without loss of generality, suppose the algorithm allocates g_1 to A and g_2 to B . Then no matter how g_3 is allocated, EF1 fails.

(i) *Allocate g_3 to agent A .* The final allocation is $\mathcal{A}_A = \{g_1, g_3\}$ and $\mathcal{A}_B = \{g_2\}$. From agent B 's perspective,

$$v_B(\mathcal{A}_A \setminus \{g_1\}) = v_B(\{g_3\}) = 100 > v_B(\mathcal{A}_B) = 1, \quad v_B(\mathcal{A}_A \setminus \{g_3\}) = v_B(\{g_1\}) = 2 > 1.$$

Hence removing either item from $\mathcal{A}_A$ does not eliminate B 's envy, so EF1 fails.

(ii) *Allocate g_3 to agent B .* The final allocation is $\mathcal{A}_A = \{g_1\}$ and $\mathcal{A}_B = \{g_2, g_3\}$. From agent A 's perspective,

$$v_A(\mathcal{A}_B \setminus \{g_2\}) = v_A(\{g_3\}) = 100 > v_A(\mathcal{A}_A) = \varepsilon, \quad v_A(\mathcal{A}_B \setminus \{g_3\}) = v_A(\{g_2\}) = 2\varepsilon > \varepsilon.$$

Again, removing either item from $\mathcal{A}_B$ does not eliminate A 's envy, so EF1 fails. $\square$

Online fair allocation naturally evokes the objective of designing online algorithms with as small a competitive ratio as possible. However, Yang et al. [35] and Gao et al. [23] show that when item information is chosen by a fully adversarial input, the competitive ratio can be as bad as $\Omega(T)$. Since the online setting typically assumes that the total number of rounds T is much larger than the number of agents and the number of item types, such a guarantee is clearly impractical. One reasonable alternative is therefore to weaken the online information model, for example by assuming that items or values are sampled from probability distributions.

In addition to online algorithm design and competitive analysis, recent work also studies no-regret online learning algorithms that aim to optimize social welfare subject to envy-related constraints, comparing the algorithm's welfare with the best fixed solution in hindsight [32]. However, that notion of envy is *ex ante* fairness. For example, when each arriving item is assigned uniformly at random, all agents have the same expected utility, much as in divisible-goods analyses. Yet a simple example in Table 1 shows that this does not guarantee *ex post* fairness: in that example, the expected maximum envy equals $\frac{1}{4}(3 + 1 + 1 + 3) = 2 > 0$. We therefore view the sublinear average-envy notion of Benadè et al. [9] as a more reasonable requirement when the goal is ex-post fairness.

$t = 1$	$t = 2$	
item value = 2	item value = 1	(a_1, a_2) envy
(0, 2)	(0, 3)	3
(0, 2)	(1, 2)	1
(2, 0)	(2, 1)	1
(2, 0)	(3, 0)	3

Table 1: Two agents. The cumulative allocations after times $t = 1, 2$ and the resulting maximum envy values. The two agents have identical valuations for goods, and the realized values received are written as (a_1, a_2) . In each round the item is allocated uniformly at random; each row represents one possible realized allocation.

1.2 Important Definitions

Inspired by the maximum envy ratio introduced by Lipton et al. [26], we use the following notions of approximate fairness (Definitions 1–4). Note that when $\alpha = 1$, α -approximate fairness reduces to the original fairness condition.

Definition 1 (α -EF). Let $\alpha \in [0, 1]$. Suppose $\mathcal{A} = (A_1, \dots, A_n)$ is an allocation of an indivisible-goods set M to agents N . If for every pair of agents $i, j \in N$,

$$v_i(A_i) \geq \alpha \cdot v_i(A_j),$$

then we call $\mathcal{A}$ an **α -approximate envy-free allocation** (**α -EF allocation**).

Definition 2 (EF1). Suppose $\mathcal{A} = (A_1, \dots, A_n)$ is an allocation of an indivisible-goods set M to agents N . If for every pair of agents $i, j \in N$,

$$\text{there exists a good } g \in A_j \text{ such that } v_i(A_i) \geq v_i(A_j \setminus \{g\}),$$

then we call $\mathcal{A}$ an **envy-free up to one good allocation** (**EF1 allocation**).

Definition 3 (α -EFX). Let $\alpha \in (0, 1]$. Suppose $\mathcal{A} = (A_1, \dots, A_n)$ is an allocation of an indivisible-goods set M to agents N . If for every pair of agents $i, j \in N$,

$$\text{for every good } g \in A_j, \quad v_i(A_i) \geq \alpha \cdot v_i(A_j \setminus \{g\}),$$

then we call $\mathcal{A}$ an **α -approximate envy-free up to any good allocation** (**α -EFX allocation**).

Definition 4 (α -MMS). Let $\alpha \in (0, 1]$. Suppose $\mathcal{A} = (A_1, \dots, A_n)$ is an allocation of an indivisible-goods set M to agents N . If for every agent $i \in N$,

$$v_i(A_i) \geq \alpha \cdot \text{MMS}_i^n(M),$$

then we call $\mathcal{A}$ an **α -approximate maximin-share fair allocation** (**α -MMS allocation**), where $\text{MMS}_i^n(M)$ is agent i 's maximin-share value $\mu_i(M)$, i.e.,

$$\mu_i(M) = \max_{B \in \mathcal{A}_i(M)} \min_{S \in B} v_i(S).$$

For the analysis and design of fair-allocation algorithms for indivisible goods, comprehensive survey articles are already available; see [3, 6]. In this proposal we focus only on research directions that are especially relevant to our goals, such as online fair allocation.

Recent frontier results suggest that without future information (for example, the aggregate future value of all goods to each agent over T rounds, or the probability distribution over future item types), the competitive ratio achievable for online fair allocation cannot in general be reduced to a constant. This project aims to unify several research threads by systematically analyzing the trade-off between fairness and efficiency (social welfare) under α -EF, α -EFX, and α -MMS guarantees. We will compare combinatorial algorithms with no-regret online-learning algorithms, which is one of the central methodological themes of this proposal.

When there is predictive information about future items with some level of accuracy, we will investigate whether online learning algorithms for indivisible-goods allocation can significantly improve expected ex-post fairness and the regret of final social welfare, and how such improvements depend on prediction accuracy. This direction is inspired by the recent NeurIPS 2024 work of Raman and Tewari [34] on online classification with predictions. Their work studies how future-item predictions can improve regret relative to hindsight, in the spirit of consistency (better performance when predictions are accurate) and robustness (performance no worse than prediction-free baselines when predictions are inaccurate), much as in the theory of algorithms with predictions.

2 Related Research Progress

2.1 Fairness Notions and Offline Fair Allocation of Indivisible Goods

2.1.1 Envy-Freeness (EF) and Envy-Freeness up to One Good (EF1)

We know that, given a set of m indivisible goods and n agents, an envy-free allocation need not exist, and deciding whether such an allocation exists is **NP**-complete [12]. Consequently, much of the literature studies relaxations of envy-freeness. The most basic and widely attainable such relaxation is Envy-Freeness up to One Good (EF1).

An allocation is *envy-free up to one good* (EF1) if any envy an agent has toward another can be eliminated by removing at most one item from the latter’s bundle. It is well known that EF1 allocations always exist for additive valuations. Polynomial-time algorithms such as round robin and envy-cycle elimination produce EF1 allocations [15, 26]. Moreover, with suitable tie-breaking or post-processing, these algorithms can also guarantee Pareto optimality (PO)⁴.

2.1.2 Envy-Freeness up to Any Good (EFX)

An allocation is *envy-free up to any good* (EFX) if, for every agent, after removing *any* one item from another agent’s bundle, no agent envies that bundle. In the general case—especially when

⁴Maximum Nash welfare allocations also simultaneously guarantee PO and EF1 [16].

there are four or more agents—the existence of EFX allocations remains a challenging open problem [15]. Manurangsi and Suksompong [28] showed, via random sampling of instances, that the probability that EFX does not exist tends to zero regardless of the relationship between the number of goods m and the number of agents n . Nevertheless, positive existence and algorithmic results are already known for several *special cases*:

- when agents have identical valuations or ordered valuations;
- when there are only two or three agents (the three-agent case admits a pseudo-polynomial-time algorithm).

Moreover, approximation methods yield α -EFX allocations in various settings (for example, $\frac{1}{2}$ -EFX, further improved to approximately 0.618-EFX [5]). More recently, Amanatidis et al. [4] proved that for additive valuations, a $\frac{2}{3}$ -EFX allocation always exists when there are at most seven agents, when agent values for goods take only three possible values, or when valuations can be represented by a multigraph (see Figure 1).

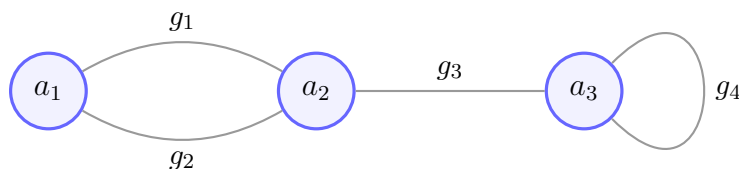


Figure 1: Illustration of a multigraph valuation instance. Vertices represent agents a_i , and edges represent goods g_j . If agent a_i values good g_j positively, then a_i is an endpoint of g_j . Goods g_1 and g_2 form parallel edges, while g_4 is a self-loop.

It is also worth noting that recent work studies approximate EFX allocations after donating a sublinear number of goods to charity [1, 17]. Roughly speaking, if one allows $\tilde{O}((n/\varepsilon)^{1/2})$ goods to remain unallocated as charity, then one can obtain a $(1 - \varepsilon)$ -EFX allocation. This is a very interesting direction for eventually resolving the existence of EFX for indivisible goods: the fewer the donated goods and the higher the approximation ratio, the closer one gets to an ideal EFX allocation. Overall, EFX-related problems have indeed become a very active research area in recent years.

2.1.3 Maximin Share (MMS)

An allocation is *maximin-share fair* (MMS) if each agent receives a bundle worth at least what she could guarantee herself by partitioning all goods into n bundles and taking the least valued one (see Definition 4). When there are more than two agents, MMS allocations are not guaranteed to exist. Furthermore, even computing the MMS value itself is NP-hard [24]. Consequently, the literature focuses primarily on approximation algorithms:

- polynomial-time algorithms can guarantee a $\frac{1}{2}$ -MMS allocation;
- later improvements achieve $\frac{2}{3}$ -MMS, and in some cases the approximation can be improved to about $\frac{3}{4} + \frac{1}{12n}$.

2.1.4 Maximum Nash Social Welfare (Maximum NSW)

Definition 5 (NSW). *Let M be the set of goods and N the set of n agents. The Nash social welfare of an allocation A is defined as the geometric mean of agents' utilities:*

$$NSW(A) = \left(\prod_{i=1}^n v_i(A_i) \right)^{1/n},$$

where v_i is agent i 's valuation function and A_i is the bundle assigned to agent i .

One can also maximize the logarithmic form $\sum_{i=1}^n \log v_i(A_i)$, which is often more numerically stable. An MNW (maximum Nash welfare) allocation maximizes the product of agents' utilities. Such allocations are noteworthy because they are both EF1 and Pareto optimal [15]. In contrast, maximizing utilitarian social welfare $\sum_{i=1}^n v_i(A_i)$ may result in some agent receiving utility zero, whereas MNW avoids such outcomes because any zero utility collapses the product to zero.

Despite their appealing fairness and efficiency properties, exactly computing MNW allocations is NP-hard. Even approximating MNW is computationally challenging, although pseudo-polynomial-time algorithms and positive results in restricted settings (such as bi-valued instances) are known. For inapproximability results, see [30].

2.2 Online Fair Allocation of Indivisible Goods

In the absence of any assumptions or restrictions on future values, online algorithms for indivisible goods generally cannot achieve bounded competitive ratios for either maximum envy or Nash social welfare under adversarial inputs [35]. We therefore focus on no-regret algorithms for online allocation. Below we briefly review two frontier results especially relevant to the goals of this project.

2.3 Honor Among Bandits (NeurIPS 2024)

2.3.1 Honor Among Bandits: No-Regret Learning for Online Fair Division [32]

Procaccia et al. [32] study the following model. There are K types of goods. When an item of type k arrives, agent i 's value is i.i.d. with unknown mean μ_{ik}^* . In each round, the algorithm chooses a *mixed* allocation $x_k \in \Delta^n$, i.e., a probability distribution over agents for that type. If type k appears with frequency p_k , then the expected utilitarian social welfare under a stationary policy $(x_k)_{k=1}^K$ is

$$USW(x; \mu^*) = \sum_{k=1}^K p_k \sum_{i=1}^n x_{ik} \mu_{ik}^*.$$

The fairness notion in that work is *ex ante fairness in each round*. For example, for envy-freeness-in-expectation (EFE), a natural per-type constraint is

$$\forall k, \forall i, j : \quad \mathbb{E}[v_i(A_i^t) \mid k] - \mathbb{E}[v_i(A_j^t) \mid k] = x_{ik}\mu_{ik}^* - x_{jk}\mu_{ik}^* \geq 0,$$

where A_i^t is the bundle agent i has after round t .

This work formalizes expected fairness (which can be represented in matrix form) and optimizes utilitarian social welfare under such fairness constraints. The algorithm learns the valuation distributions by sampling and optimizes expected welfare under ex-ante fairness. In some sense, the problem resembles a multi-armed bandit problem, where each variable x_{ik} can be viewed as an arm. Their final result gives a regret bound of $\tilde{O}(T^{2/3})$, compared against the expected utilitarian welfare of the best fixed allocation distribution in hindsight that satisfies the ex-ante fairness constraints over all T rounds. For more details, see also my presentation slides: https://josephcclin.github.io/paper/honor_among_bandits_nips2024.pdf.

2.3.2 How to Make Envy Vanish Over Time [9] (EC 2018): Ex-Post Fairness via Vanishing Envy

Benadè et al. [9] consider the following setting: items arrive one by one and must be allocated immediately; allocations are irrevocable. Let A_i be agent i 's bundle after T rounds, and define the *ex-post* maximum envy by

$$\text{Envy}_T = \max_{i,j} [v_i(A_j) - v_i(A_i)]_+.$$

The goal is to make the *average* envy Envy_T/T tend to zero (vanishing envy). They proved that a simple *uniform random allocation* rule—assign each arriving item to a uniformly random agent—achieves

$$\mathbb{E}[\text{Envy}_T] \in \tilde{O}(\sqrt{T \log T / n}),$$

and hence $\mathbb{E}[\text{Envy}_T]/T \rightarrow 0$ as $T \rightarrow \infty$.

2.4 Guarantees of EFX Allocations of Indivisible Goods for Maximizing NSW

EFX with charity [14]. For additive valuations, Caragiannis, Gravin, and Huang [14] proved that starting from an allocation maximizing NSW, by donating (i.e., leaving unallocated) some items, one can obtain a *partial* EFX allocation⁵ whose NSW is at least one half of the optimal NSW, and this factor $\frac{1}{2}$ is tight. Moreover, if one starts from a ρ -approximate maximum-NSW allocation, their algorithm guarantees an EFX allocation whose NSW is at least $\frac{1}{2\rho}$ times the optimum. They also provide a lower bound showing that no EFX (partial) allocation can guarantee more than $\frac{1}{2} + o(1)$ of the optimal NSW.

⁵It is called “partial” because not all goods are allocated.

Trade-off between α -EFX and maximum Nash social welfare [20]. Let $\alpha \in [0, 1]$. Feldman, Murras, and Ponitka [20] characterized the **optimal joint guarantee** between approximate EFX and MNW. For additive valuations, every instance admits a partial α -EFX allocation whose NSW is at least a $\frac{1}{1+\alpha}$ fraction of the maximum Nash welfare. This trade-off is tight. When $\alpha \leq \varphi - 1 \approx 0.618$, they further show that one can transform the partial allocation into a *complete* allocation without weakening either guarantee. Algorithmically, if an input allocation X already achieves at least a β fraction of the maximum NSW, then their procedure outputs an α -EFX allocation Y with NSW at least $\frac{\beta}{1+\alpha}$ times optimum. Thus, their method converts approximate maximum-NSW allocations into approximate EFX allocations efficiently.

- (II) Research methods, implementation steps, and execution schedule. Please describe them year by year: 1. the research methods adopted in this project, the reasons for choosing them, and their innovation; 2. anticipated difficulties and possible solutions; 3. coordination and use of important instruments; 4. if overseas research is required, please explain its necessity and expected benefits.

1 Year 1

Question 1. *In the first year, we consider the following two goals:*

Design and analysis of online no-regret algorithms for maximizing social welfare under α -approximate envy-freeness and α -approximate envy-freeness up to one good (α -EF or α -EF1) in stationary or dynamic environments.

Design and analysis of online no-regret algorithms for maximizing social welfare under the vanishing-envy condition in stationary or dynamic environments.

Remark. The result of Benadè et al. [9] shows that uniform random allocation guarantees vanishing envy, namely $\mathbb{E}[\text{Envy}_T] = o(T)$. However, it remains worthwhile to study whether this algorithm, or other online allocation algorithms with vanishing envy, can also provide sufficiently strong guarantees for utilitarian social welfare (USW) or Nash social welfare (NSW).

1.1 A Promising Machine-Learning Starting Point: A Gradient-Based Iterative Update Algorithm

Let N be the set of n agents and M the set of m types of indivisible goods. We define $\mathbf{X}(t) \in \mathbb{R}^{n \times m}$, $\mathbf{V}(t) \in \mathbb{R}^{n \times m}$, and $\mathbf{U}(t) \in \mathbb{R}^{n \times n}$ as follows. Let $\mathbf{X}^{(t)} = (x_{ij}^{(t)})_{i \in [n], j \in [m]}$ denote the allocation at time t , where $x_{ij}^{(t)}$ represents the fraction of item j assigned to agent i . Let $\mathbf{V}^{(t)} = (v_{ij}^{(t)})_{i \in [n], j \in [m]}$ denote agent i 's value for item j at time t , where $t = 1, \dots, T$. By matrix multiplication we obtain

$$\mathbf{X}^{(t)} \mathbf{V}^{(t)\top} = \mathbf{U}^{(t)},$$

where $\mathbf{U}^{(t)} = (u_{ij}^{(t)})_{i, j \in [n]}$. In particular, $u_{ii}^{(t)}$ is the utility received by agent i (see Figure 2).

This matrix-based formulation is convenient for GPU-accelerated implementation of learning algorithms. Several fairness-related quantities can be expressed naturally in this framework:

- the *minimum share* of allocation $\mathbf{X}^{(t)}$ is

$$\text{MinShare}(\mathbf{X}^{(t)}) = \min_{i \in [n]} u_{ii}^{(t)};$$

- the *maximum envy* of allocation $\mathbf{X}^{(t)}$ is

$$\text{MaxEnvy}(\mathbf{X}^{(t)}) = \max \left(0, \max_{i \in [n]} \max_{j \in [n]} (u_{ij}^{(t)} - u_{ii}^{(t)}) \right);$$

$$\begin{bmatrix} x_{11}^{(t)} & x_{12}^{(t)} & \cdots & x_{1m}^{(t)} \\ x_{21}^{(t)} & x_{22}^{(t)} & \cdots & x_{2m}^{(t)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1}^{(t)} & x_{n2}^{(t)} & \cdots & x_{nm}^{(t)} \end{bmatrix} \times \begin{bmatrix} v_{11}^{(t)} & v_{21}^{(t)} & \cdots & v_{m1}^{(t)} \\ v_{12}^{(t)} & v_{22}^{(t)} & \cdots & v_{m2}^{(t)} \\ \vdots & \vdots & \ddots & \vdots \\ v_{1n}^{(t)} & v_{2n}^{(t)} & \cdots & v_{mn}^{(t)} \end{bmatrix} \longrightarrow \begin{bmatrix} u_{11}^{(t)} & u_{12}^{(t)} & \cdots & u_{1n}^{(t)} \\ u_{21}^{(t)} & u_{22}^{(t)} & \cdots & u_{2n}^{(t)} \\ \vdots & \vdots & \ddots & \vdots \\ u_{n1}^{(t)} & u_{n2}^{(t)} & \cdots & u_{nn}^{(t)} \end{bmatrix}$$

$\mathbf{X}^{(t)} \qquad \qquad \mathbf{V}^{(t)\top} \qquad \qquad \mathbf{U}^{(t)}$

Figure 2: Mapping from the allocation matrix and valuation matrix to the utilities received by agents.

- the *Nash social welfare* of allocation $\mathbf{X}^{(t)}$ is

$$\text{NSW}(\mathbf{X}^{(t)}) = \left(\prod_{i=1}^n u_{ii}^{(t)} \right)^{1/n}.$$

To achieve fairness, one may therefore try, at each round $t \in [T]$ or after the final round T , to maximize $\text{MinShare}(\mathbf{X}^{(t)})$, minimize $\text{MaxEnvy}(\mathbf{X}^{(t)})$, or maximize $\text{NSW}(\mathbf{X}^{(t)})$.

Since we consider indivisible goods, we define the feasible solution space by

$$\mathcal{Q} = \left\{ \mathbf{X} \in \mathbb{N}^{n \times m} : x_{ij} \in \{0, 1\} \ \forall i \in [n], j \in [m] \text{ and } \sum_{i \in [n]} x_{ij} = 1 \ \forall j \in [m] \right\}.$$

A simple and intuitive implementation of the projection mapping $\text{Proj}_{\mathcal{Q}}(\mathbf{X}^{(t)})$ is:

- (i) normalize each column by setting $\hat{x}_{ij}^{(t)} = x_{ij}^{(t)} / \sum_{i=1}^n x_{ij}^{(t)}$ for all $i \in [n], j \in [m]$; then
- (ii) sample an agent for item j according to the probability distribution $(\hat{x}_{ij}^{(t)})_{i \in [n]}$ (in Python, this can be implemented via `random.choice()`).

We may consider different loss functions $f : \mathbf{X}^{(t)} \times \mathbf{V}^{(t)} \rightarrow W^{(t)}$, where f can be instantiated as $-\text{MinShare}(\mathbf{X}^{(t)})$, $\text{MaxEnvy}(\mathbf{X}^{(t)})$, or $-\text{NSW}(\mathbf{X}^{(t)})$, depending on the fairness objective. This yields the following general gradient-based iterative update algorithm (Algorithm 1). Note that $W^{(t)}$ need not be differentiable everywhere, so $\frac{dW^{(t)}}{d\mathbf{X}} \in \mathbb{R}^{n \times m}$ may be a subgradient, as in [7].

We conjecture that the gradient-based framework, when equipped with suitable loss functions, may lead to algorithms achieving both vanishing envy and large Nash social welfare.

Algorithm 1: A General Gradient-Based Algorithm for Online Fair Allocation

- 1: Set $\eta > 0$ to be a sufficiently small constant.
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: **Compute** $\mathbf{U}^{(t)} = \mathbf{X}^{(t)} \mathbf{V}^{(t)\top}$ and $W^{(t)} = f(\mathbf{U}^{(t)})$.
 - 4: **Update** $\mathbf{Y}^{(t+1)} \leftarrow \mathbf{X}^{(t)} - \eta \frac{dW^{(t)}}{d\mathbf{X}}$.
 - 5: **Project** $\mathbf{Y}^{t+1}$ to be feasible: $\mathbf{X}^{(t+1)} \leftarrow \text{Proj}_{\mathcal{Q}}(\mathbf{Y}^{t+1})$.
 - 6: **end for**
-

2 Year 2

Question 2. *In the second year, we consider the following two goals:*

- *Assuming the existence of predictions of each agent's total value for future goods (or the arrival probabilities of goods in the next round), design and analyze online no-regret algorithms for maximizing social welfare under α -approximate envy-freeness and α -approximate envy-freeness up to one good (α -EF or α -EF1) with a predictor.*
- *Assuming the existence of predictions of each agent's total value for future goods (or the arrival probabilities of goods in the next round), design and analyze online no-regret algorithms for maximizing social welfare under the vanishing-envy condition with a predictor.*

2.1 Envy Graphs and Envy-Cycle Elimination

Envy graphs are widely used in the literature on offline envy-free allocation. Although the main problem in this project is online, we believe that graph-theoretic analysis through envy graphs and related variants can help us better understand algorithm design and correctness. We use the following simple example to introduce the standard envy graph. Suppose there are three goods x, y, z and three agents with values:

	x	y	z
Agent 1	6	2	1
Agent 2	3	5	2
Agent 3	2	1	6

Assume the initial allocation is $A_1 = \{y\}$, $A_2 = \{z\}$, and $A_3 = \{x\}$, and all valuations are additive. Then we obtain the envy graph shown in Figure 3: vertices represent agents, and there is a directed edge from i to j if agent i envies agent j 's bundle. By applying Algorithm 3, we obtain a graph with no envy cycle, corresponding to an EF1 allocation.

It is important to note that in Algorithm 2, even after all cycles are eliminated, some envy edges may remain; that is, the graph may still be a forest. One must therefore choose a source

vertex (an agent with in-degree zero), and if there are still unallocated goods, assign such a good to the corresponding agent. If a new envy cycle is created, Algorithm 3 is used again. This framework first appeared in the work of Lipton et al. [26] and guarantees EF1. We propose to use it as a baseline while seeking higher social welfare⁶.

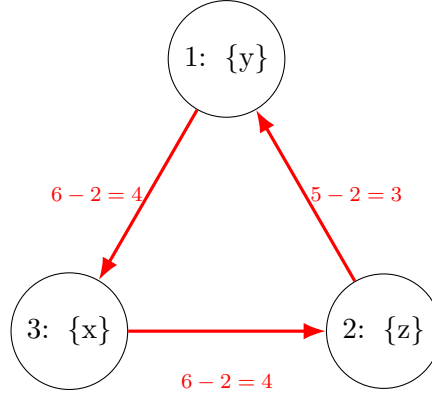


Figure 3: Envy graph for three agents. The red edges form the envy cycle $1 \rightarrow 3 \rightarrow 2 \rightarrow 1$; the numbers on the edges are the corresponding envy values $u_i(A_j) - u_i(A_i)$.

Algorithm 2: Envy-Cycle Elimination (ECE) for Additive Valuations

Input: Agents $N = \{1, 2, \dots, n\}$; goods G ; additive valuations $v_i : G \rightarrow \mathbb{R}_{\geq 0}$

Output: Allocation $A = (A_1, A_2, \dots, A_n)$

for each $i \in N$ **do**

$A_i \leftarrow \emptyset$

while $G \neq \emptyset$ **do**

 pick any $g \in G$

 // unallocated good

 build envy graph $D(A)$ with vertices N and edge $i \rightarrow j$ iff $v_i(A_i) < v_i(A_j)$

$S \leftarrow$ sources of $D(A)$

 // agents with no incoming edge

 choose $r \in \arg \max_{i \in S} v_i(\{g\})$

$A_r \leftarrow A_r \cup \{g\}; \quad G \leftarrow G \setminus \{g\}$

while $D(A)$ contains a directed cycle C **do**

 ROTATECYCLE(A, C)

 update $D(A)$

Return A

⁶As mentioned earlier, any allocation maximizing Nash social welfare is EF1, but an EF1 allocation need not maximize Nash social welfare.

Algorithm 3: ROTATECYCLE($A, C = (i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_k \rightarrow i_1)$)

```

 $B \leftarrow A_{i_1}$ 
for  $t \leftarrow 1$  to  $k - 1$  do
     $A_{i_t} \leftarrow A_{i_{t+1}}$ 
 $A_{i_k} \leftarrow B$ 
// each agent receives the bundle previously held by the agent she
   envied

```

2.2 A Possible Framework for Achieving Consistency and Robustness via Predictions

Inspired by the machine-learning-augmented algorithmic design paradigm of Lykouris and Vassilvitskii [27], and by the prediction-augmented no-regret learning framework considered by Raman and Tewari [34], we propose the following initial framework as a starting point for this project.

Given an online instance S of length T , let **ALG** be an online algorithm processing S . Suppose that **ALG** has access to a prediction object $\tilde{\Theta}$ (for example, the total future value $\tilde{V}$ of all future goods to all agents, or the predicted arrival frequencies $\tilde{p}$ of goods). Let $\text{OPT}(S)$ denote the chosen offline hindsight benchmark. Let $L_T(\text{ALG}; S)$ denote the loss of the algorithm (or minus the objective value), and define regret by

$$R_T(\text{ALG} \mid \tilde{\Theta}; S) := L_T(\text{ALG}; S) - L_T(\text{OPT}; S).$$

Fix a nonnegative prediction-error functional $\text{Err}_T(S, \tilde{\Theta})$, so that $\text{Err}_T = 0$ under perfect prediction. Let B_T denote a baseline robust upper bound achievable in the worst case without prediction.

Definition 6 (Consistency and Robustness of Algorithms with Predictions). *An online algorithm with predictions $\text{ALG}_{\tilde{\Theta}}$ is called:*

(C) consistent (with respect to Err_T) if there exists a nondecreasing function $\phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ with $\phi(0) = 0$ and constants $c_0, c_1 \geq 0$ such that

$$R_T(\text{ALG}_{\tilde{\Theta}} \mid \tilde{\Theta}; S) \leq c_1 \phi(\text{Err}_T(S, \tilde{\Theta})) + c_0 \quad \text{for all } S.$$

In particular, under perfect prediction ($\text{Err}_T = 0$), the regret is $O(1)$.

(R) robust if there exist constants $C_0, C_1 \geq 0$ such that for all S and all $\tilde{\Theta}$,

$$R_T(\text{ALG}_{\tilde{\Theta}} \mid \tilde{\Theta}; S) \leq C_1 B_T + C_0.$$

*That is, asymptotically the algorithm never performs worse than a no-prediction baseline. A stronger **best-of-both-worlds** (BoBW) guarantee is*

$$R_T(\text{ALG}_{\tilde{\Theta}} \mid \tilde{\Theta}; S) \leq \min\{C_1 B_T + C_0, c_1 \phi(\text{Err}_T) + c_0\}.$$

Possible error measures include

$$\text{Err}_{\text{tot}} = \|\tilde{V} - V\|_{\infty}, \quad \text{Err}_{\text{freq}} = \|\tilde{p} - p\|_{\text{TV}}, \quad \text{Err}_{\text{ctx}} = \frac{1}{T} \sum_{t=1}^T \max_i |\tilde{v}_{i,t} - v_{i,t}|.$$

Here

$$\|\tilde{p} - p\|_{\text{TV}} := d_{\text{TV}}(\tilde{p}, p) = \frac{1}{2} \sum_{k=1}^K |\tilde{p}_k - p_k| = \frac{1}{2} \|\tilde{p} - p\|_1,$$

which is the total variation distance between two probability distributions p and $\tilde{p}$.

For example, consider log-NSW regret together with ex-post envy. Let $\text{LNSW}(A) = \sum_{i=1}^n \log(1 + U_i)$ and $R_T^{\text{LNSW}} = \text{LNSW}^* - \text{LNSW}(A^T)$. Let $E_{\max}(T) = \max_{t \leq T} \max_{i,j} [v_i(X_j^t) - v_i(X_i^t)]_+$. Fix a baseline envy envelope $\mathcal{E}_T$ (for example, $\mathcal{E}_T = O(\sqrt{\log T})$ under uniformly random arrivals).

Definition 7 (Consistency and Robustness for log-NSW and Fairness). *A prediction-augmented online allocator satisfies*

$$\textbf{Consistency:} \quad \mathbb{E}[R_T^{\text{LNSW}}] \leq \phi(\text{Err}_T) + o_T(1), \quad \phi(0) = 0,$$

and also

$$\textbf{Robustness (BoBW):} \quad \mathbb{E}[R_T^{\text{LNSW}}] \leq CB_T \quad \text{and} \quad E_{\max}(T) \leq \mathcal{E}_T \quad (\text{ex post, on the realized sequence,})$$

A combined target statement is

$$\mathbb{E}[R_T^{\text{LNSW}}] \leq \min\{CB_T, \phi(\text{Err}_T)\} \quad \text{and} \quad E_{\max}(T) \leq \mathcal{E}_T.$$

If one uses predictions of total future values or future item frequencies, then natural calibrated targets are

$$\mathbb{E}[R_T^{\text{LNSW}}] \leq \tilde{O}(\sqrt{T}) + C_1 \|\tilde{V} - V\|_{\infty} \quad \text{or} \quad \mathbb{E}[R_T^{\text{LNSW}}] \leq \tilde{O}(\sqrt{T}) + C_2 \|\tilde{p} - p\|_{\text{TV}}.$$

The $\tilde{O}(\sqrt{T})$ term yields robustness, while the additive penalty vanishes under perfect prediction, giving consistency. On the fairness side, the same ex-post envelope (for example, on maximum envy) can be maintained regardless of prediction quality.

3 Year 3

Question 3. *In the third year, we aim to explore sufficient conditions for the existence of EFX allocations. Concretely:*

Assume that in each round we may decide whether the current item is donated to charity, and that after the final T rounds the value of the charity bundle to every agent is no greater than the value of that agent's own allocation, while the number of charity items is required to be $o(T)$. Does there exist an online allocation algorithm that computes an α -approximate EFX allocation? Equivalently, does there exist a constant value of α ?

Even in the offline setting, this is a difficult and active problem. We also note that the online setting resembles the classical secretary problem [21]: deciding immediately whether to donate an arriving item to charity is analogous to deciding immediately whether to hire the current candidate, and our problem corresponds to hiring at most $k = o(T)$ secretaries [11].

A Simple Preliminary Observation on EFX

We begin with a warm-up observation about offline EFX allocations. Without loss of generality, suppose each agent has nonnegative value for each good. Then if the number of agents n is at least the number of items m , an EFX allocation always exists.

Proposition 1. *Given a set M of m indivisible goods and a set N of n agents, if $n \geq m$, then there exists an EFX allocation.*

Proof. Choose any subset $N' \subseteq N$ of size m . Assign exactly one item from M to each agent in N' . Then for any pair of agents $i, j \in N$, if i envies j , agent j can remove the item she received, after which i no longer envies j . $\square$

From Proposition 1, we see that in the online setting, nontrivial EFX difficulty can arise only when $T > n$. Moreover, the following observation shows that once some items are fixed in advance, an instance that originally admitted an EFX allocation may become one that does not.

Observation 2 (cf. [10]). *Suppose $m = 5$, $n = 2$, and both agents value the five goods as 1, 1, 3, 4, 9. If the first agent receives the four goods of values 1, 1, 3, 4 and the second receives the good of value 9, then the allocation is clearly envy-free. However, if initially each agent is given one of the two goods of value 1, then regardless of how the remaining three goods are allocated, EFX cannot be achieved: the agent holding the good of value 9 would have to discard that good, otherwise the other agent would envy her.*

We may also use the recent AAAI 2025 result of Feldman et al. [20] as a reference point. They study the optimal trade-off between EFX fairness and Nash social welfare (NSW), asking

whether there exists an allocation satisfying both α -EFX and β -NSW. They prove that for any $\alpha \in [0, 1]$, there exists a *partial* allocation satisfying α -EFX and EF1 whose NSW is at least $1/(\alpha + 1)$ of the optimum. They further show that when $\alpha < 0.618$, the partial allocation can be transformed into a complete allocation while preserving the same fairness and efficiency guarantees. For subadditive valuations, they show that when $\alpha \leq 1/2$, there exists a complete α -EFX allocation with NSW at least $1/(\alpha + 1)$ times optimal.

If time permits, we would like to attack the longstanding open problem of the existence of EFX allocations for indivisible goods. Our intuition is that the degree of EFX violation may be captured by a suitable potential function.

Question 4. *Does there exist a dynamic-update algorithm for reallocating indivisible goods such that the algorithm measures the degree of EFX violation by a potential function, and reaching potential at most zero certifies that an EFX allocation has been found?*

Does there exist a dynamic-update algorithm for reallocating indivisible goods such that the algorithm measures the degree of EFX violation by a potential function, and reaching potential at most zero certifies that an EFX allocation has been found?

Preliminary ideas for potential functions. We may consider the following candidate potentials for designing dynamic algorithms that decrease the potential and eventually obtain an EFX allocation:

1. a direct quantity from the definition of EFX:

$$\Delta_{i \rightarrow j}(A) := \max \left\{ 0, v_i(A_j) - \min_{g \in A_j} v_i(g) - v_i(A_i) \right\}, \quad \forall i \neq j;$$

2. an ℓ_1 -type aggregate envy potential:

$$\Phi_1(A) := \sum_{i \neq j} \Delta_{i \rightarrow j}(A);$$

3. an ℓ_∞ -type maximum envy potential:

$$\Phi_\infty(A) := \max_{i \neq j} \Delta_{i \rightarrow j}(A);$$

4. the number of ordered agent pairs violating EFX (also related to Branzei's work [13]):

$$\Phi_{\text{count}}(A) := |\{(i, j) \in [n]^2 : i \neq j, \Delta_{i \rightarrow j}(A) > 0\}|.$$

The same potential-based ideas may also be applied to the design of online allocation algorithms for indivisible goods.

(III) Expected tasks and outcomes. Please describe year by year: 1. the expected tasks to be completed; 2. the training expected for project personnel; 3. expected research outputs (such as practical impact, journal papers, conference papers, books, technical reports, patents, or technology transfer, including both quantity and quality); 4. expected contributions to academic research, national development, and other applications.

1 Expected Tasks to Be Completed

Year 1

- Collect, study, and discuss the literature on offline and online fair allocation.
- Investigate and implement algorithms for online approximate envy-free allocation, and analyze how approximate fairness affects utilitarian and Nash social welfare.
- Prepare manuscripts and submit them to international conferences or journals.

Year 2

- Collect, study, and discuss literature on learning- or prediction-augmented algorithms.
- Investigate how learning or prediction augmentation can enhance algorithms for online approximate envy-free allocation.
- Analyze the effectiveness and robustness of learning- or prediction-augmented algorithms.
- Prepare manuscripts and submit them to international conferences or journals.

Year 3

- Collect, study, and discuss the literature on EFX allocations.
- Understand the extremal-graph-theoretic guarantees behind the rainbow cycle number [1, 17] and investigate more refined analyses.
- Explore whether fewer charity goods can lead to stronger approximate EFX guarantees, together with associated algorithmic designs.
- Investigate appropriate potential functions and feasible dynamic-update algorithms for decreasing them, in an attempt to prove the existence of EFX allocations.
- Prepare manuscripts and submit them to international conferences or journals.

2 Expected Training for Project Personnel

- Improved ability to search for and read research literature.
- Stronger skills in algorithm design, analysis, and implementation.
- Familiarity with game theory and price-of-anarchy analysis.
- Understanding of current research trends in international game theory, computational social choice, machine learning theory, and related AI topics.
- Familiarity with development environments such as Anaconda, Jupyter Notebook, Spyder, and PyCharm.
- Practical ability to implement machine-learning algorithms in Python/PyTorch.
- Improved skills in report and paper writing.
- Improved ability in planning and managing research progress.
- Improved ability to collaborate and divide work in solving research problems.

3 Expected Research Outputs

The academic outputs of this project are expected to be submitted to major international conferences and SCI journals in the relevant fields.

The international conferences we target include:

- ACM Conference on Economics and Computation (EC).
- International Conference on Machine Learning (ICML).
- Annual Conference on Neural Information Processing Systems (NeurIPS).
- International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS).
- Association for the Advancement of Artificial Intelligence (AAAI).
- International Joint Conferences on Artificial Intelligence (IJCAI).
- ACM-SIAM Symposium on Discrete Algorithms (SODA).
- International Symposium on Algorithmic Game Theory (SAGT).
- International Symposium on Algorithms and Computation (ISAAC).
- Conference on Web and Internet Economics (WINE).
- International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM).
- Games, Agents, and Incentives Workshop (GAIW).
- International Workshop on Computational Social Choice (COMSOC).

- Annual Meeting of the Asian Association for Algorithms and Computation (AAAC).
- IJCAI Workshop on Computational Fair Division.

The international journals we consider include:

- Games and Economic Behavior (GEB).
- Artificial Intelligence (AIJ).
- International Journal of Game Theory (IJGT).
- Autonomous Agents and Multi-Agent Systems (JAAMAS).
- Algorithmica.
- Theoretical Computer Science (TCS).
- Journal of Computer and System Sciences (JCSS).
- Theory of Computing Systems (TOCS).
- Social Choice and Welfare (SCW).

4 Expected Contributions to Academic Research, National Development, and Other Applications

1. Advances in theoretical methodology

- Establish a two-objective analytical framework combining **ex-post fairness (vanishing envy)** and **social welfare (USW/NSW)** for online indivisible-goods allocation, yielding **sublinear envy bounds** and **sublinear regret bounds** for log-NSW/USW.
- Develop prediction-augmented online allocation algorithms for indivisible goods with **best-of-both-worlds (BoBW)** guarantees: **consistency** when the prediction error is small, and **robustness** in the worst case.
- Explore an online version of **EFX-with-charity**: design a **sublinear-size charity pool** as a buffer mechanism, with the goal of achieving $(1 - \varepsilon)$ -EFX while donating only $B(T) = o(T)$ goods overall.

Quantitative targets: submit 1–2 papers per year to top conferences such as EC/NeurIPS/ICML/AAMAS and complete 3–4 SCI/SCIE journal papers within three years (for example, GEB, AIJ, Algorithmica, TCS, JAAMAS, and SCW).

2. Academic community and talent cultivation

- Train graduate students in **algorithm design, analysis, and implementation**, and in interdisciplinary topics spanning **game theory, fair allocation, and online learning**.
- Organize reading groups with domestic and international scholars, and pursue further collaborative projects.

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